

Python Quick Documentation

A concise introduction to Python programming.

1. What is Python?

Python is a high-level, interpreted, and general-purpose programming language known for its readability and extensive ecosystem of libraries.

2. Basic Syntax

Variables: `x = 10`

Strings: `name = 'Alice'`

Lists: `nums = [1, 2, 3]`

Conditionals: `if x > 5: print(x)`

Loops: `for n in nums: print(n)`

3. Functions

Functions are defined using the 'def' keyword.

Example: `def greet(name): return f'Hello {name}'`

4. Object-Oriented Programming

Python supports classes and objects for building reusable and maintainable code.

5. Popular Libraries

NumPy (numerical computing), Pandas (data analysis), Matplotlib (visualization), FastAPI (APIs), and Django (web development).

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